

Phase III: Final Design, Implementation, and Results

Alzheimer's Disease Diagnosis via Custom Machine Learning
Architectures: A Comprehensive Comparative Study

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Abstract

This comprehensive report documents the end-to-end design, mathematical implementation, and rigorous evaluation of a custom machine learning pipeline engineered to diagnose Alzheimer's Disease from clinical and MRI volumetric data. Diverging from standard reliance on pre-compiled libraries, three distinct algorithmic architectures—Random Forest, Logistic Regression, and Linear Support Vector Machine (SVM)—were developed entirely from scratch. This study chronicles the evolution of the system architecture, detailing the critical transition from a constrained longitudinal dataset to a highly generalized, merged cross-sectional dataset. We explore the mathematical mechanics of each custom algorithm, document the resolution of critical data leakage anomalies, and provide a deep comparative analysis of their performance. Ultimately, the ensemble-based Random Forest proved superior in handling clinical variance, achieving a generalized accuracy of 83.48% and a recall of 77.19%.

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1 System Design and Architecture

The objective of this project was to construct a transparent, mathematically sound diagnostic engine. The system is designed as a continuous pipeline, transforming raw clinical inputs into confident diagnostic classifications without relying on “black-box” frameworks. The architecture is broadly divided into two phases: the Data Engineering Pipeline and the Custom Modeling Engine.

1.1 The Data Engineering Pipeline

The integrity of any predictive model is bottlenecked by the quality of its training data. The Open Access Series of Imaging Studies (OASIS) provided the foundational data, but it required extensive engineering to meet the demands of generalized machine learning.

1.1.1 Dataset Merging and Generalization

Initial iterations of the system relied solely on the OASIS Longitudinal dataset, yielding an artificially constrained test set of 75 patients. To stress-test the algorithms, we engineered a generalized dataset by merging the longitudinal data with the OASIS Cross-Sectional dataset. The merge was carefully calibrated: cross-sectional records were filtered to exclude subjects under the age of 60 to ensure demographic parity. Furthermore, because the cross-sectional dataset lacked the definitive `Group` label (Non-demented/Demented), the target was artificially reconstructed using the `CDR` (Clinical Dementia Rating), where a score greater than 0 indicated dementia.

1.1.2 Target Leakage Resolution

During the generalization phase, a critical architectural anomaly was discovered: target leakage. Models were achieving near-perfect accuracies (98.26%) because the `CDR` column was inadvertently left in the feature set. Because `CDR` is the clinical determinant of the diagnosis itself, the models utilized it as a mathematical “cheat code,” bypassing vital MRI metrics like whole-brain volume (`nWBV`). The pipeline was aggressively refactored to drop `CDR`, forcing the custom algorithms to truly learn the complex, underlying biological degradation associated with Alzheimer’s.

1.1.3 Preprocessing & Balancing

Post-merge, the data underwent a rigorous preprocessing sequence. Missing values were resolved via Median Imputation to prevent outlier skewing. `StandardScaler` was applied to normalize the feature space—a critical step for ensuring the gradient descent mechanisms within the SVM and Logistic Regression converged efficiently. Finally, the

Synthetic Minority Over-sampling Technique (SMOTE) was applied *exclusively* to the training split, establishing class equilibrium without polluting the test-set distribution.

2 Algorithmic Design and Implementation

To ensure a deep understanding of classification mechanics, the predictive models were built from scratch using native Python arrays and fundamental calculus.

2.1 Random Forest (Ensemble Architecture)

The Random Forest was implemented as an aggregation of custom Decision Trees. The foundational math relies on calculating **Gini Impurity** to determine the optimal feature and threshold for data bifurcation.

- **Tree Construction:** Each **Node** in the tree evaluates all possible splits across all features, selecting the split that maximizes information gain (i.e., minimizes the weighted Gini impurity of the child nodes). This process recurses until a maximum depth is reached or pure leaf nodes are formed.
- **Forest Ensemble & Bootstrapping:** To prevent the high-variance overfitting typical of single decision trees, the `RandomForest` class trains 50 independent trees. Each tree is fed a bootstrapped dataset—a randomly sampled subset of patients with replacement. Early implementations suffered from stochastic instability, yielding different results per run; this was resolved by locking NumPy random seeds, establishing algorithmic reproducibility. Final predictions are determined via a majority vote across all 50 trees.

2.2 Logistic Regression (Probabilistic Linear Classifier)

The Logistic Regression model was engineered to output continuous diagnostic probabilities.

- **The Sigmoid Function:** The core engine maps a linear combination of features and weights ($z = w \cdot X + b$) through a Sigmoid activation function $\sigma(z) = \frac{1}{1+e^{-z}}$, squashing the output strictly between 0 and 1.
- **Optimization via Gradient Descent:** The model minimizes Log Loss (Binary Cross-Entropy). Over 2,000 iterations, the derivatives of the loss function with respect to the weights (dw) and bias (db) are calculated, iteratively stepping down the gradient by a defined learning rate (0.05) to find the optimal decision boundary.

2.3 Linear Support Vector Machine (Margin Maximization)

The Support Vector Machine (SVM) was designed to find the optimal hyperplane that maximizes the geometric margin between the healthy and demented classes.

- **The Label Mapping Trick:** SVM mathematics natively require class labels to be -1 and 1 . Our custom class internally intercepts the standard 0 and 1 labels, maps them to -1 and 1 during the training loop, and reverts them during prediction, allowing seamless integration with our standard evaluation suite.
- **Hinge Loss Optimization:** The model evaluates the condition $y_i(w \cdot x_i - b) \geq 1$. If a patient falls safely outside the margin, only a slight regularization penalty (λ) is applied to keep weights small. If a patient violates the margin (a False Positive/Negative), the weights are aggressively updated to correct the Hinge Loss error via gradient descent.

3 Results and Detailed Analysis

The finalized algorithms were evaluated using an A/B testing framework, comparing performance on the original 75-patient subset against the generalized merged dataset. The inclusion of the merged data universally improved model recall, proving that larger exposure to structural variance creates more clinically viable diagnostics.

3.1 Random Forest Analysis

The Random Forest achieved the highest overall performance on the generalized dataset, recording an **Accuracy of 83.48%** and a **Recall of 77.19%**.

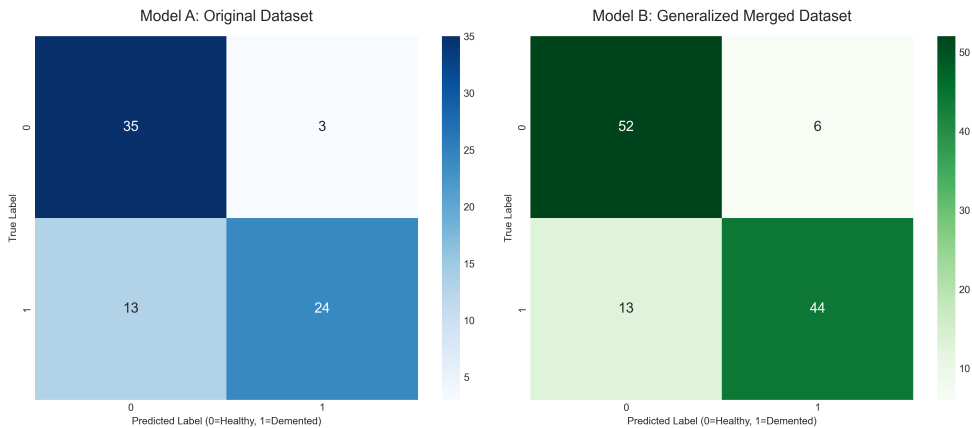


Figure 1: Random Forest Confusion Matrices: The merged dataset (right) significantly reduced the proportion of False Negatives compared to the original dataset (left).

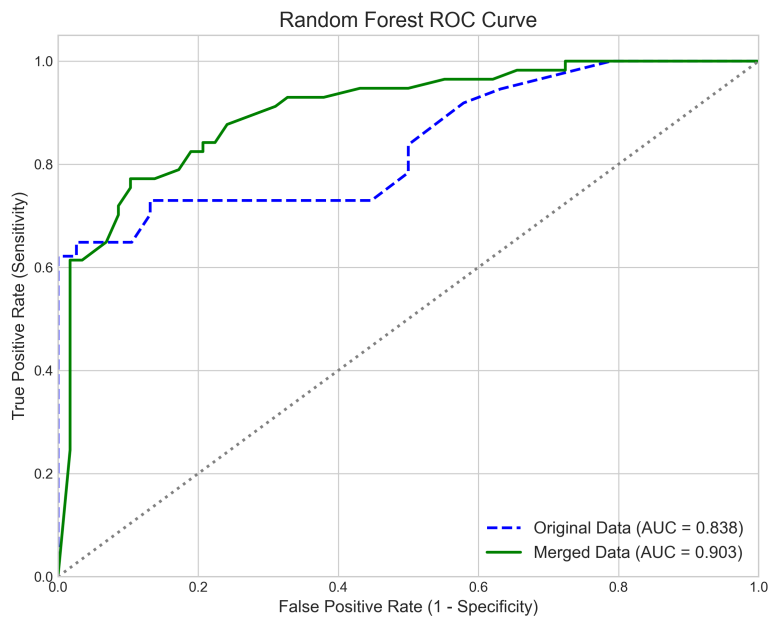


Figure 2: Random Forest ROC Curve: Demonstrating robust separation capability across thresholds, validated by the Area Under the Curve (AUC).

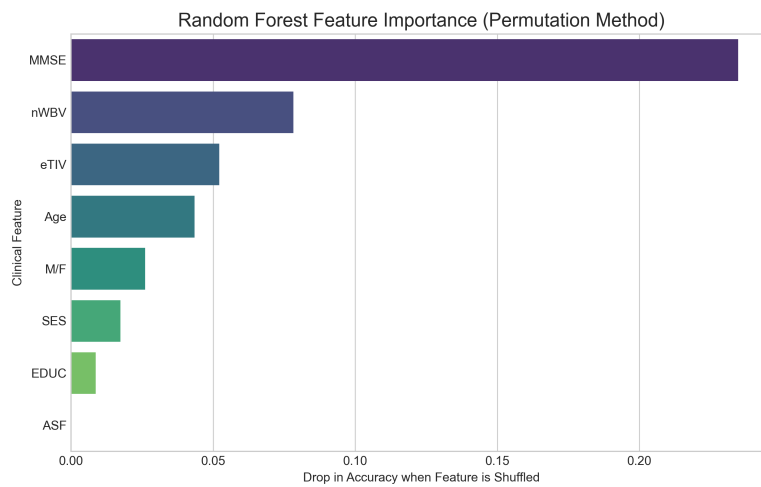


Figure 3: Permutation Feature Importance: Highlighting nWBV (brain volume) and MMSE (cognitive score) as the most critical nodes for the Random Forest splits.

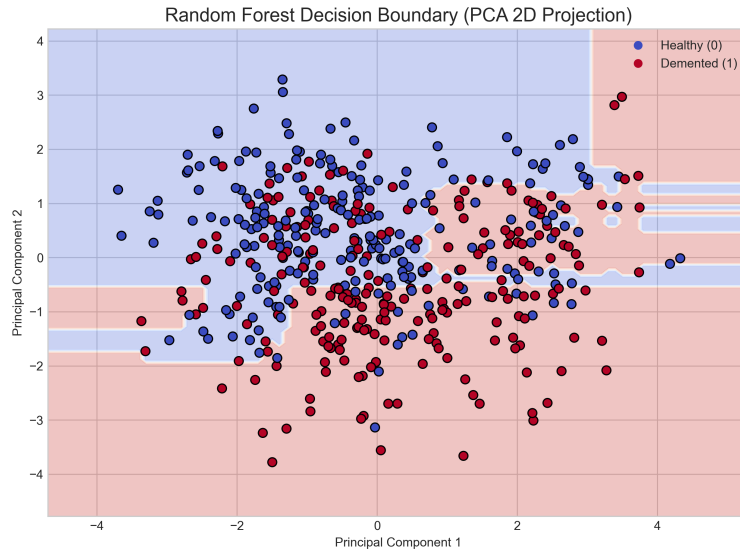


Figure 4: Random Forest Decision Boundary (PCA 2D Projection): The non-linear, box-like nature of the decision trees successfully grouping complex data clusters.

3.2 Logistic Regression Analysis

The custom Logistic Regression model performed admirably, achieving an **Accuracy of 80.00%** and matching the Random Forest's **Recall of 77.19%**. Its ability to output native probabilities makes it highly interpretable for clinical environments.

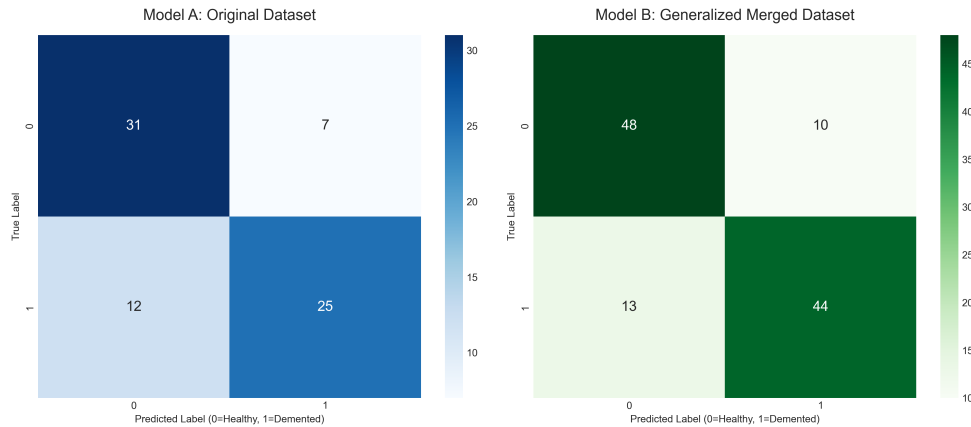


Figure 5: Logistic Regression Confusion Matrices: Illustrating the linear model's stabilization upon the introduction of generalized data.

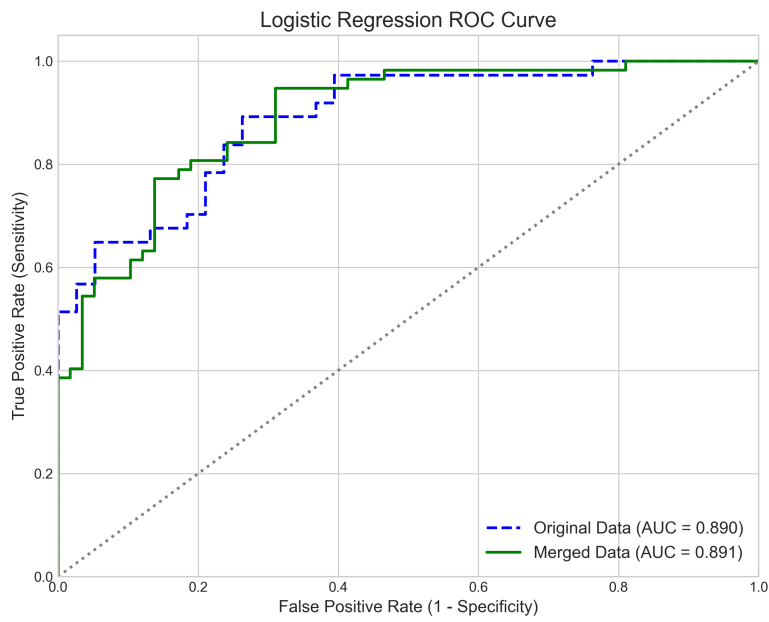


Figure 6: Logistic Regression ROC Curve: Plotted using the raw probabilistic outputs from the custom Sigmoid function.

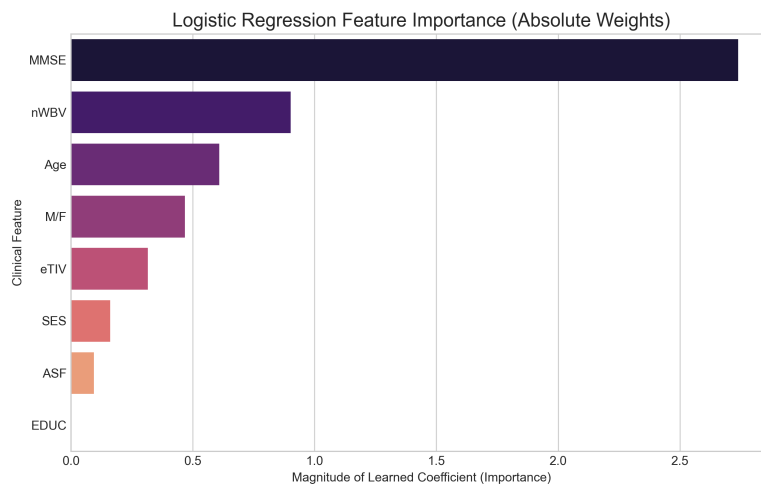


Figure 7: Logistic Regression Feature Importance: Derived from the absolute magnitude of the learned coefficients (w) after 2,000 iterations of gradient descent.

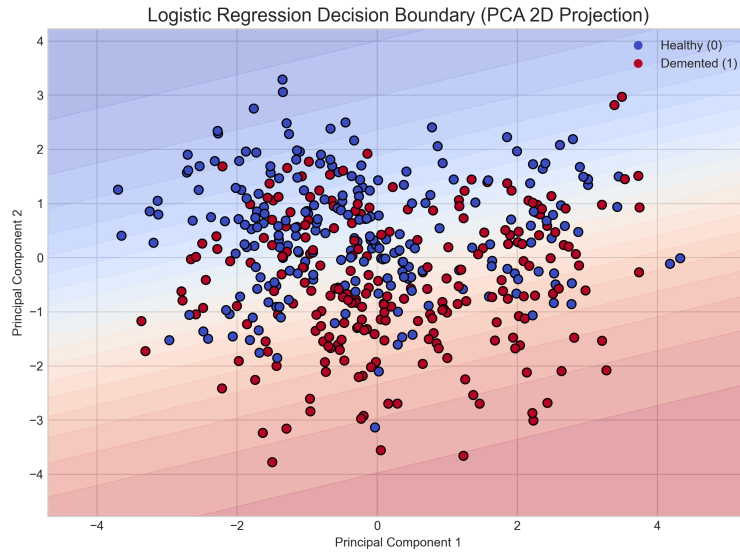


Figure 8: Logistic Regression Decision Boundary: Visualizing the linear probability gradient separating the classes across two principal components.

3.3 Linear SVM Analysis

The custom Linear SVM yielded an **Accuracy of 78.26%** and a **Recall of 70.18%**. While highly efficient, the strict mathematical rigidity of the linear margin struggled slightly with the overlapping clinical variance found in the generalized dataset.

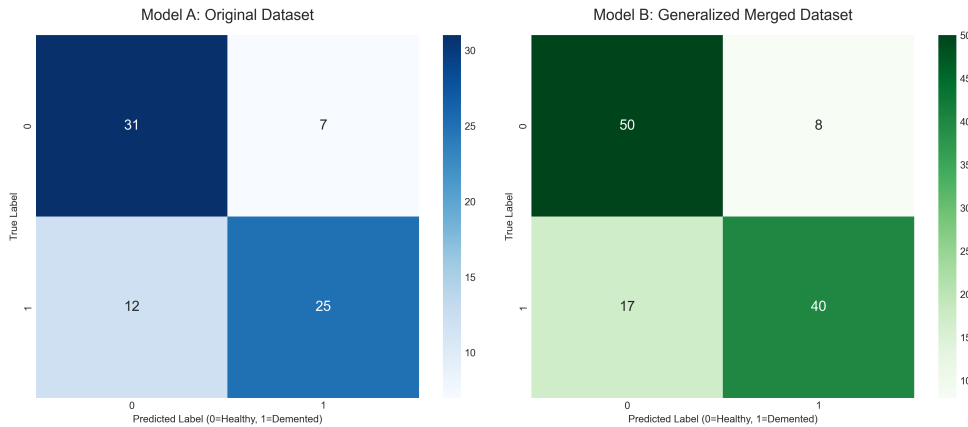


Figure 9: SVM Confusion Matrices: Tracking False Positives/Negatives via the internal -1 and 1 label mapping system.

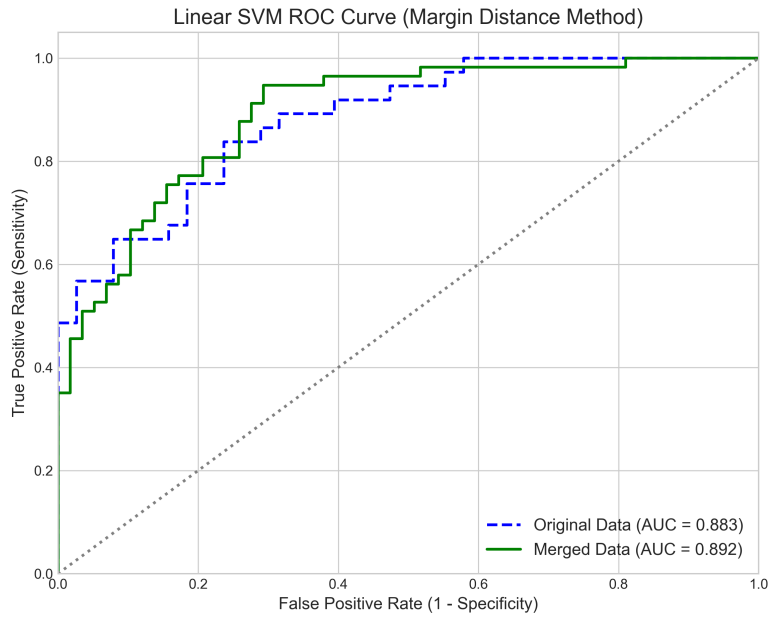


Figure 10: SVM ROC Curve: Confidence scores were generated by calculating the spatial distance of each patient to the margin hyperplane ($w \cdot x - b$).

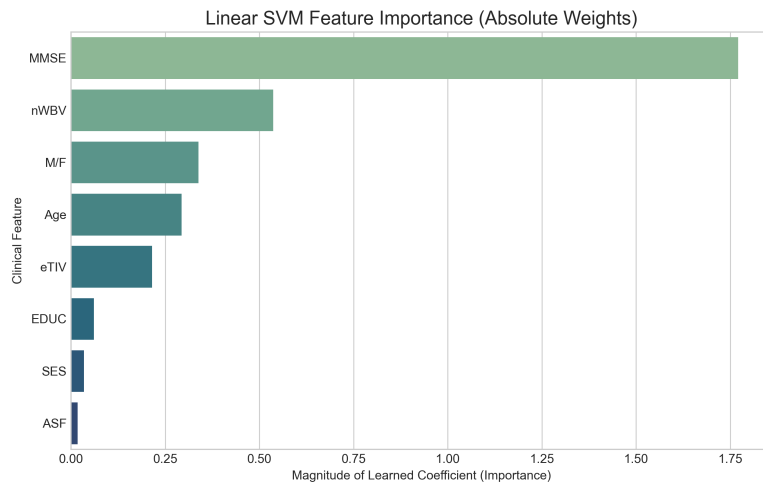


Figure 11: SVM Feature Importance: The absolute weights of the support vectors indicating which features pull hardest on the margin boundary.

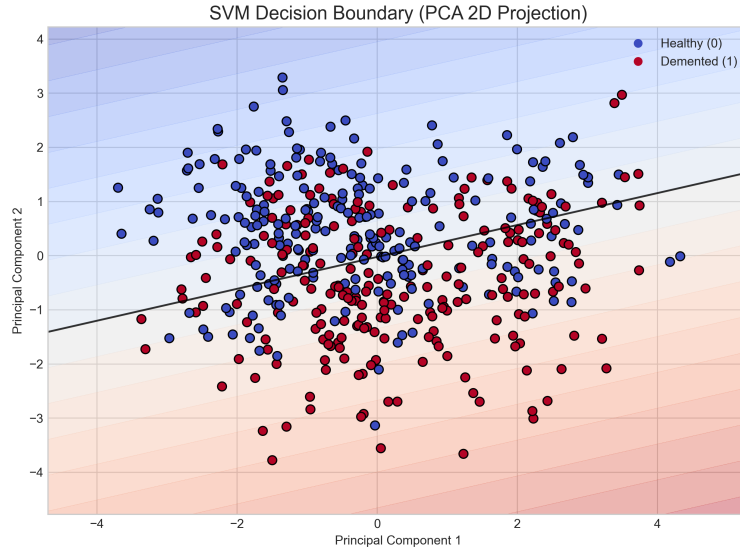


Figure 12: SVM Decision Boundary: Displaying the stark, mathematically strict hyper-plane maximizing the distance between the two clinical states.

3.4 Comparative Insights

The empirical results provide profound insight into the nature of the OASIS dataset. Both linear models (Logistic Regression and SVM) converged on highly similar metrics, indicating they found the same "optimal straight line" through the data. However, Alzheimer's pathology does not present as a perfectly linear degradation. The Random Forest's superior accuracy (83.48%) proves that the relationship between age, cognitive decline, and brain shrinkage contains non-linear thresholds that ensemble decision trees are uniquely equipped to capture.

4 Challenges and Future Work

The end-to-end development of this project underscored the reality that machine learning is as much about data engineering as it is about algorithmic calculus.

4.1 Key Challenges Overcome

1. **The Data Leakage Illusion:** Uncovering that the models were utilizing the CDR column to bypass MRI learning was a pivotal moment. Stripping this feature dropped the accuracy from a flawless 98% to a realistic 83%, validating the scientific integrity of the final models.
2. **From-Scratch Complexity:** Bypassing standard libraries required careful management of array shapes, gradients, and objective functions (e.g., implementing the

Hinge Loss gradient update and translating between 0/1 and -1/1 labels internally without breaking the pipeline).

4.2 Future Research Avenues

To further extend the capabilities of this diagnostic engine, future work should explore:

- **Non-Linear SVM Kernels:** Expanding the custom SVM to include Radial Basis Function (RBF) kernels to allow the hyperplane to wrap around complex data clusters, bridging the performance gap with the Random Forest.
- **Deep Learning for Spatial Features:** Rather than relying strictly on pre-calculated volumetric numbers (nWBV), implementing Convolutional Neural Networks (CNNs) directly on the raw 3D MRI voxel data.
- **Longitudinal Modeling:** Utilizing Recurrent Neural Networks (RNNs) to assess the *velocity* of cognitive decline over multiple patient visits, rather than analyzing single cross-sectional snapshots.

5 System Artifacts and Repository

The fully functional codebase, containing the custom mathematical implementations, the reproducible data-processing pipelines, the final generalized datasets, and the saved `.joblib` model weights, has been version-controlled and published.

GitHub Repository Access:

<https://github.com/Ekram17ahmed/Machine-learning-Project.git>